

T1 – Controlled Current Source

Analog Bench Instrument – Constant Current Load, 0–10 A / 42 V

Wave 1 | Status: **RED – Not Started** | Layers: 2 | MCU: None | CAN: None

1 Purpose

Bench test instrument for validating PULSAR power electronics. Provides a continuously adjustable constant-current load from 0–10 A at up to 42 V. Used to: test Power Module output regulation under load, verify Power Master OCP trip thresholds, stress-test buck converters, and characterize power path resistance. Entirely analog – no MCU, no firmware.

2 Topology

N-channel MOSFET operating in the linear (ohmic-to-saturation) region as a variable resistor. A negative-feedback loop sets the drain current:

1. **Setpoint:** 10-turn potentiometer generates a 0–3.3 V reference voltage.
2. **Current Sense:** A low-side shunt resistor (R_{shunt}) converts drain current to a voltage: $V_{\text{sense}} = I_D \times R_{\text{shunt}}$.
3. **Sense Amplifier:** INA293A1 (20 V/V gain) amplifies V_{sense} to a feedback voltage.
4. **Error Amplifier:** MCP6001 compares the feedback to the setpoint. Output drives the NMOS gate.
5. **NMOS:** Gate voltage adjusts to maintain $V_{\text{feedback}} = V_{\text{setpoint}}$, thus regulating I_D .

$$\text{Current setpoint: } I_{\text{set}} = \frac{V_{\text{pot}}}{G_{\text{INA}} \times R_{\text{shunt}}} = \frac{V_{\text{pot}}}{20 \times 0.01} = V_{\text{pot}} \times 5 \text{ A/V}.$$

At $V_{\text{pot}} = 2.0 \text{ V} \Rightarrow I_{\text{set}} = 10 \text{ A}$ (full scale).

3 Key Components

Function	Part	Package	Notes
Current Sense Amp	INA293A1 (std)	SOT-23-5	20 V/V, 0–110 V CM, STRICT UNIDIRECTIONAL
Error Amplifier	MCP6001 (std)	SOT-23-5	Rail-to-rail, 1 MHz GBW
Shunt Resistor	10 m Ω , 5 W	2512 Std	Kelvin-style layout required
NMOS Pass Element	Power N-MOSFET	TO-220 / TO-247	PCB-mounted, Low $R_{\text{DS(on)}}$
Setpoint Pot	10 k Ω 10-turn	Panel-mount	Bourns 3590S or equivalent
I/O Connectors	XT30 Horizontal Male	PCB-mount	Rated $\geq 30 \text{ A}$ (XT30 limit check)
Data Readout	2-pin Header	2.54 mm	Buffered output (Oscilloscope)
Ext. Feedback	3-pin Header	2.54 mm	Option for external INA module

3.1 NMOS Pass Element Requirements

$V_{\text{DS(max)}}$	$\geq 60 \text{ V}$ (42 V system + margin)
$I_{\text{D(cont)}}$	$\geq 15 \text{ A}$ at $T_C = 100^\circ\text{C}$
$R_{\text{DS(on)}}$	Low $R_{\text{DS(on)}}$ preferred (minimize voltage drop/heat)
$P_{\text{D(max)}}$	$\geq 100 \text{ W}$ (with heatsink)
$V_{\text{GS(th)}}$	2–4 V (driven by 3.3 V op-amp, consider level-shift if needed)
SOA	Must handle high $V_{\text{DS}} \times I_{\text{D}}$ simultaneously
Package	TO-220 or TO-247 (PCB-mounted)

4 Thermal Design

Max Power Dissipation	$V_{\text{DS}} \times I_{\text{D}} = 42 \text{ V} \times 10 \text{ A} = 420 \text{ W}$ (absolute max)
Sustained Rating (target)	200 W continuous
Heatsink	PCB-mounted with Integrated 12V Fan (powered by PCB)
θ_{JA} Target	$\leq 0.5^\circ\text{C/W}$ (junction to ambient, with heatsink + fan)
Thermal Protection	NTC on heatsink, comparator trips at 80°C , disables gate drive

Note: At 420 W the MOSFET SOA and heatsink are at their limits. Recommend limiting continuous use to 200 W. For higher power, parallel a second NMOS with matched gate drive.

5 Electrical Specifications

Input Voltage Range	0–42 V (applied by DUT or bench supply)
Current Range	0–10 A continuously adjustable
Current Resolution	~5 mA (10-turn pot, 2000 turns over 10 A)
Current Accuracy	±1% (limited by shunt tolerance and INA293 gain error)
Power Dissipation	Up to 420 W peak, 200 W sustained
Control	Manual (10-turn pot), no digital interface
Aux Supply	5 V bench supply for op-amp and sense amp (<10 mA)

6 Schematic Review Checklist

- ☐ INA293A1: Shunt Kelvin-style routing (pseudo-Kelvin traces)
- ☐ INA293A1: REF pin to GND, OUT to MCP6001 inverting input + Data Readout Header
- ☐ INA293A1: Provision for external INA module (input jumper/header)
- ☐ MCP6001: non-inverting input from potentiometer wiper
- ☐ MCP6001: output drives NMOS gate (check $V_{GS(th)}$)
- ☐ MCP6001: compensation cap (10–100 pF) across feedback path
- ☐ NMOS gate: series resistor (10–100 Ω)
- ☐ NMOS: gate-source Zener clamp (≤ 20 V)
- ☐ Potentiometer: 100 nF bypass cap on wiper
- ☐ Aux 5 V supply: 100 nF + 10 μ F decoupling
- ☐ XT30 Connectors: Correct polarity, footprint check
- ☐ Thermal protection: NTC near NMOS + comparator circuit
- ☐ Fan Control: 12V supply for Heatsink Fan
- ☐ Power traces: ≥ 5 mm width for 10 A path
- ☐ NMOS PCB footprint: Verify TO-220/247 pinout and spacing for heatsink

7 BOM Readiness Checklist

- ☐ INA293A1 – in Component Standards, sourced
- ☐ MCP6001 – in Component Standards, sourced
- ☐ Shunt resistor (10 m Ω , 5 W, 2512) – Standard footprint
- ☐ NMOS – Low $R_{DS(on)}$, TO-220/247, PCB-mountable
- ☐ 10-turn potentiometer – part selected (panel-mount, 10 k Ω)
- ☐ XT30 Connectors – Male Horizontal
- ☐ Heatsink + Fan – PCB mountable, 12V fan
- ☐ Headers – 2.54mm for Readout and Ext. Feedback
- ☐ NTC thermistor – part selected
- ☐ Compensation capacitor value identified
- ☐ Gate resistor value identified
- ☐ Aux 5 V / 12 V supply connectors specified
- ☐ All passives in 0603/0805 standard values
- ☐ No single-source components
- ☐ All lead times <2 weeks